Patrick Mockridge

Aberystwyth, Wales · patrickmockridge@gmail.com · patrickmockridge.github.io

PROFILE AND METHOD

Independent protocol engineer and formal-verification specialist. Process safety taught me that a plausible number is not a safe number — so every result here is either proved or carries its limits. Process safety engineer, 2006–2015; chartered 2013; independent protocol and verification work since 2016.

The method: specify the failure mode before the architecture; prove what can be proved; contain what cannot; record provenance with every result. DarkWow is containment; rnode-rust is continuity; Azoth is provenance; Mudra is verification in a familiar form.

SELECTED WORK

DarkWow

Designed for private organisations on a public chain, without an administrator or a public voter roll

2026–; redesign April–October 2026 · Research prototype; Lean-modelled opcodes/gadgets (partial; 8 axioms; OBL-Z12 open); manual circuit audit; HAZOP tabletop

rnode-rust

A Rust node for the ρ-calculus: native object capabilities and machine-checked laws

2026; port August–October 2026 · Pre-release; 72/77 laws; 19 laws (23 entries) node-tied; 14 crates, 0 unsafe

Azoth

A process calculation library where every number carries its provenance

2026 · v0.1.x; 94 upstream issues; 217 provenance tests; not for design work

Mudra

Prove a committed calculation trace is unchanged without revealing the spreadsheet

2026 · Research build; ~6 KB proof; 21 field elements; 100–632-instruction demos

Records: patrickmockridge.github.io/darkwow/ · /rnode-rust/ · /azoth/ · /mudra/

Engagements: protocol review, rescue, and verification; a bounded triage with written findings, then a scoped remediation. Remote from Aberystwyth, Wales.

CHRONOLOGY

PERIOD	WORK
2026	Delivery: rnode-rust port (August–October), DarkWow redesign (April–October), Azoth, Mudra.
2022–2026	Independent research and writing: network analysis, evidence archiving, influence/secrecy simulation; RGov/Rhobot recovery proposal for REV/RChain with Jim Whitescarver (2024); UNISS/Ingenesist simulations (2026).
2016–2019	Treehouse AI and Robotics software nanodegrees; early ERC-1155 context at Enjin (2016); named advisor, Enjin Coin ICO (2017); CoEngineers, early investor (2017–2019).
2006–2015	Chemical engineer, process safety; UK and North Africa. Chartered 2013. Bitcoin since 2011; London Ethereum community, 2014.
2002–2006	MEng Chemical Engineering, University of Edinburgh.

RESEARCH AND WRITING

Romeo–Juliet critical risk-analysis framework; UNISS/Ingenesist simulations. Selected writing · Ingenesist archive.